

Autonomous inverse modeling of complex groundwater systems via a physics-integrated large language model multi-agent framework

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HIGHLIGHTS

- Hydro-Agent integrates LLMs with physics engines for autonomous inverse modeling.
- Autonomous agents generate and debug scripts for rigorous process-based simulators.
- The code-as-policy paradigm enforces strict thermodynamic consistency and transparency.
- This approach resolves mass balance violations inherent in deep learning surrogates.

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ABSTRACT

Inverse modeling of groundwater flow and transport in subsurface systems is fundamentally restricted by the conceptual-numerical gap, where translating hydrogeological hypotheses into executable simulation codes remains a labor-intensive process. While data-driven machine learning offers superior computational speed, it frequently violates mass conservation laws and lacks the interpretability required for regulatory decision-making. To resolve these operational bottlenecks, we propose Hydro-Agent, a physics-integrated multi-agent framework that leverages Large Language Models (LLMs) to govern rigorous process-based simulations via a “code-as-policy” paradigm. Unlike black-box predictors, the framework couples a reasoning agent (HydroCoder) with an execution agent (Executor) to autonomously generate, debug, and adaptively configure optimization strategies for standard solvers. We systematically validate the framework’s robustness through three progressive tiers of hydrogeological complexity. Results demonstrate that the agents successfully bridge the conceptual-numerical gap: (1) In heterogeneous media, the system autonomously identified structural controls of non-convex geological features; (2) In kinetic experiments, it demonstrated robustness across multiple noise levels to achieve high-fidelity parameter retrieval under significant observational uncertainty; and (3) In the field-scale Aquia Aquifer, it navigated an 18-dimensional parameter space to reproduce complex chromatographic separation trends along a 96 km flow path while strictly enforcing mechanistic thermodynamic consistency. Systematic baseline comparisons confirm that Hydro-Agent maintains the numerical rigor of traditional direct-coupling approaches while significantly enhancing operational autonomy and ensuring geologically plausible outcomes. Hydro-Agent establishes a transparent, auditable, and physically constrained AI modeling framework. Ultimately, this abstraction of technical complexity into high-level intent broadens access for hydrogeologists who lack coding expertise, facilitating objective-driven modeling workflows that prioritize hydrogeological insights over computational implementation.

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1. Introduction

Inverse modeling of groundwater systems remains one of the most fundamental yet persistent challenges in hydrogeology. Accurate predictions of fluid flow and solute transport depend heavily on the precise characterization of subsurface properties, such as hydraulic conductivity, storage coefficients, and geochemical reaction parameters (Hill and Tiedeman, 2007). Conventional manual trial-and-error (Samper and Neuman, 1989) persists in modern practice, forcing modelers to iteratively tune parameter fields against observed hydraulic heads or concentrations. In addition to being labor-intensive, the reliance on domain intuition fails to scale with the escalating complexity of modern hydrogeological models (Zhou et al., 2014). For three-dimensional models, exploring high-dimensional parameter spaces can necessitate weeks of repeated simulations (Laloy and Vrugt, 2012). Additionally, nonlinear subsurface processes, such as multicomponent cation exchange or density-dependent flow, often produce objective function surfaces with multiple local minima, complicating the search for optimal parameters even for experienced practitioners (Mao et al., 2006).

To overcome these challenges, various optimization and statistical inversion methods have been developed. Gradient-based algorithms, including Newton, conjugate-gradient, and Levenberg–Marquardt methods (Agyei and Hatfield, 2006; Zhao et al., 2022), are computationally efficient but highly sensitive to initial conditions and prone to getting stuck in local minima (Dai and Samper, 2004; Ma et al., 2026). Bayesian inference techniques, such as Markov Chain Monte Carlo (MCMC), provide a rigorous quantification of uncertainty but require an impractically large number of forward simulations, rendering them computationally prohibitive when confronted with the high-dimensional grids of regional-scale flow models (Dai et al., 2019; Huang and Dai, 2008; Vrugt, 2016). Global metaheuristic methods, such as genetic algorithms and particle swarm optimization, broaden the search space but often require extensive hyperparameter tuning and substantial computational resources (Ketabchi and Ataie-Ashtiani, 2015). Moreover, these classical mathematical approaches are inherently limited to numerical inputs, often fail to assimilate soft data, such as descriptive logs of paleochannels or qualitative facies architectures, which carry critical spatial information that remains challenging to encode into traditional cost functions (Renard, 2007). This reliance on expert-driven coding and manual setup creates a high entry barrier and limits the scalability of these methods in real-world applications.

Concurrently, advancements in artificial intelligence (AI) have shown promise for hydrological prediction (Chen et al., 2025). Recent advances in deep learning models, such as convolutional neural networks (CNNs) (Pan et al., 2022), long short-term memory (LSTM) networks (Ng et al., 2023), and hybrid data-driven architectures (Liu and Song, 2025), have demonstrated strong predictive capabilities across various hydrological tasks. However, these models typically operate as black boxes that capture statistical correlations rather than mechanistic causality (Chen and Dai, 2024). A major limitation is their dependency on high-density labeled datasets, which are often scarce in subsurface investigations due to the sparse distribution of observation wells (Xia et al., 2023). Additionally, purely data-driven models often lack explicit grounding in physical laws. While physics-informed neural networks (PINNs) attempt to mitigate this by embedding partial differential equations into loss functions, they often suffer from convergence issues in stiff, multi-scale geological systems and lack seamless integration with industry-standard simulators (Karniadakis et al., 2021; Raissi et al., 2019). Consequently, these neural architectures offer limited interpretability, poor generalization to unseen geological conditions, and limited capacity to incorporate domain-specific mechanistic priors (Dai et al., 2025). Once trained, their parameters remain fixed, making it difficult to adapt to new hydrogeological conditions without extensive retraining (Ma et al., 2025a). These limitations underscore a critical gap: although AI is increasingly applied to hydrology, existing approaches rarely support transparent, physics-integrated reasoning or the

autonomous orchestration of numerical groundwater models.

Recent advancements in Large Language Models (LLMs) offer a potential solution to bridge this gap (Chen et al., 2024; Hadid et al., 2024; Hosseini and Pourzangbar, 2025). However, as noted in recent evaluations (Hu et al., 2025), standard stateless LLMs often struggle with the rigorous logical consistency required for long-horizon scientific workflows (Wang et al., 2024). To address this, the field is evolving toward autonomous agentic architectures that utilize memory modules and planning algorithms to decompose complex tasks (Ye et al., 2024). Although recent studies have applied such multi-agent systems to engineered domains, such as water distribution network optimization (Hedaiaty Marzouny and Dziedzic, 2024; Wang et al., 2025), urban floodwater depth estimation (Lyu et al., 2025), and hydraulic interaction (Goldshtein et al., 2025), subsurface environments present distinct challenges due to their inherent spatial heterogeneity and limited observability. To date, the potential of leveraging the semantic reasoning and code-generation capabilities of LLMs to execute rigorous, physics-based groundwater simulations for automated parameter inversion remains underexplored. This research proposes a paradigm shift: unlike surrogate models that rely on statistical approximations of physical laws, LLM-based agents can emulate expert hydrogeologists by formulating, executing, and iteratively debugging physics-based model code (Ma et al., 2025b). This method enables a new form of AI-driven, white-box modeling, where the decision process is represented by transparent code rather than opaque neural weights. By shifting the AI's role from a passive text generator to an active code executor, this framework automates high-dimensional parameter estimation while ensuring that all solutions maintain strict interpretability and physical consistency.

To capitalize on this potential, we introduce Hydro-Agent, a closed-loop, multi-agent framework capable of autonomously performing parameter inversion for complex groundwater systems. While comprehensive field modeling encompasses labor-intensive tasks such as stratigraphic geometry construction and mesh discretization, this study specifically targets the reasoning bottlenecks inherent in inverse modeling and parameter identification. The framework features a cooperative architecture centered around two core agents. The Hydro-Coder, functioning as a computational hydrogeologist, interprets natural language tasks and generates calibration scripts using standard physics engines, while dynamically refining algorithms based on execution error feedback. Working in tandem, the Executor serves as the runtime environment, responsible for dispatching simulations and providing diagnostic feedback. Through iterative interaction, these agents autonomously navigate the objective function landscape, selecting and configuring robust global optimizers to find optimal parameters even in complex, non-convex solution spaces. We validate the framework's versatility through three increasingly complex case studies: the identification of non-convex geological structures in a steady-state flow model, a high-precision inversion of nonlinear reactive transport under multiplicative noise, and a real-world, high-dimensional reactive transport model of the Aquia aquifer involving 18 coupled hydro-geochemical parameters. These case studies demonstrate that LLM-driven agents can autonomously conduct transparent, reproducible modeling workflows, establishing a reliable human-on-the-loop paradigm for navigating the high-dimensional, non-convex landscapes characteristic of real-world hydrogeology.

2. Methodology

2.1. Architecture of the physics-integrated multi-agent system

In this study, we propose Hydro-Agent, a closed-loop multi-agent framework designed to bridge the semantic reasoning capabilities of LLMs with the rigorous determinism of physics-based groundwater simulations. Unlike data-driven surrogates that attempt to map inputs directly to outputs via statistical correlations, our framework operates

on a “code-as-policy” principle (Liang et al., 2022). In this architecture, the LLM functions not as a black-box predictor, but as a transparent reasoning engine. It translates natural language intent (e.g., “minimize the RMSE of the solute breakthrough curve”) into executable Python logic, which dynamically coordinates numerical solvers to perform the computationally intensive simulations. Specifically, the framework is engineered to support a heterogeneous software environment: it seamlessly integrates both modern modular simulators and established geochemical codes regardless of their native interface protocols. This hybrid approach ensures that every step of the inversion process is transparent, mathematically verifiable, and reproducible.

As illustrated in Fig. 1, the framework operationalizes this principle through a coherent three-stage pipeline that progresses from constraint definition to iterative execution and transparent reporting. The workflow initiates with domain-knowledge prompt engineering (Fig. 1a), where the search space is rigorously bounded by physics-integrated constraints, interface protocols, and observational schemas. These inputs feed into the Hydro-Agent reasoning loop (Fig. 1b), the core computational engine. Here, the HydroCoder generates Python logic which is dispatched by the Executor to the external physics engine. Crucially, this reasoning loop features a dual-feedback mechanism designed to enforce both syntactic correctness and physical consistency. As depicted by the dashed red path in Fig. 1b, the Syntax Self-correction module acts as a defensive layer: if the Executor detects a runtime traceback or API error, it triggers an immediate rectification loop, prompting the agent to debug its code. Conversely, successful simulations trigger the solid blue path of physics-informed Optimization, where quantitative metrics (e.g., RMSE) guide the agent to refine hydrogeological parameters. Finally, to ensure auditability, the framework generates Interpretable outputs (Fig. 1c). Unlike opaque black-box models, Hydro-Agent produces comprehensive artifacts, including running audit logs for logic verification, CSV-based optimization histories for trajectory analysis, and spatially distributed simulated fields for hydrogeological validation.

2.2. Domain-knowledge embedded prompt engineering

The efficacy of the framework relies on a specialized strategy we term domain-knowledge embedded prompt engineering. Unlike standard conversational interactions where users provide vague instructions, our framework employs a structured context injection mechanism. Based on the system messages defined in our HydroCoder

agent, we formalize the conversation between the human expert and the LLM into a comprehensive constraint framework. This manifold rigorously defines the boundaries of the inversion problem, transforming the LLM from a generic text generator into a constrained reasoning engine compliant with hydrogeological principles.

2.2.1. HydroCoder: context injection for constrained reasoning

The core logic of the HydroCoder is governed by a structured system prompt, the coding_orchestrator, which constructs a multi-dimensional constraint manifold within the LLM’s context window. To effectively guide the transition from natural language intent to executable numerical code, this prompt integrates three critical categories of prior information to rigorously constrain the solution space.

First, to ensure geological plausibility, the agent is anchored within a physics-informed search space. The prompt explicitly defines the prior intervals for all constitutive parameters, thereby forcing the generated optimization routines to operate strictly within a physically plausible hypercube. This explicit bounding prevents the proposal of mathematically valid but physically unrealistic values, effectively mitigating the risk of “hallucination” (Ji et al., 2023) and ensuring consistency with site-specific hydrogeological knowledge.

Second, the prompt embeds algorithmic and interface protocols to navigate the non-convex optimization landscape. Recognizing the complexity of subsurface inversion, the agent is instructed to dynamically evaluate the suitability of optimization strategies and tune hyperparameters based on problem dimensionality, noise characteristics, and physical response smoothness. The final selection and subsequent adaptive refinements are made autonomously during runtime reasoning rather than being hard-coded in the framework. To operationalize this, we inject distinct I/O directives tailored to the specific physics engine: for modular simulators like MODFLOW (Bakker et al., 2016), the agent is directed to leverage object-oriented flopy methods to programmatically modify package arrays; conversely, for fixed-format codes like TOUGHREACT (Xu et al., 2006), it is provided with file-parsing templates to perform precise ASCII string manipulation on rigid input files. This distinction ensures that the agent can accurately map the optimizer’s decision vector to model parameters regardless of the underlying API architecture.

Finally, to mitigate context window limitations, we employ an observational metadata injection strategy. Rather than handling voluminous raw field data, the system provides only the structural schema of the observation files, such as column headers and unit conventions. This

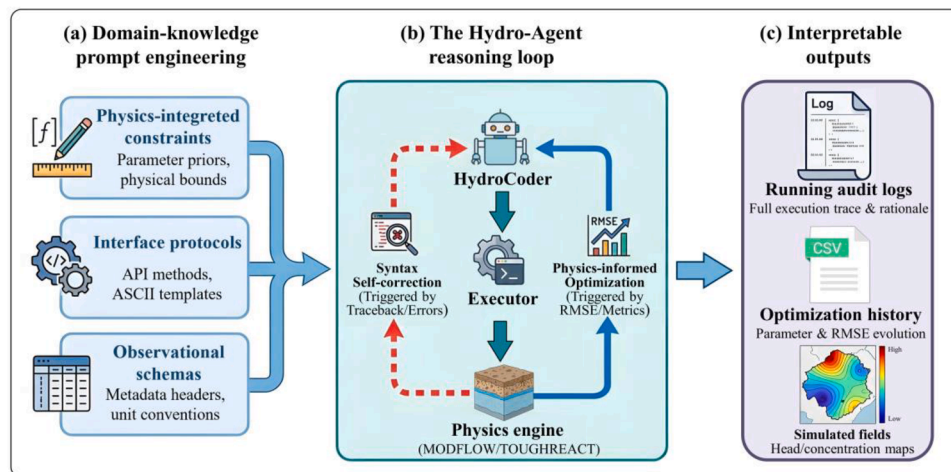


Fig. 1. The architectural workflow of the Hydro-Agent framework. (a) Domain-knowledge inputs: The context injection phase defining the search space via constraints and schemas. (b) The reasoning loop: The central iterative process featuring a dual-path feedback mechanism: Syntax self-correction (red dashed arrows) for error handling, and physics-informed optimization (blue solid arrows) for parameter refinement. (c) Interpretable outputs: Final deliverables including audit logs, optimization trajectories, and simulated fields. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

enables the agent to generate Python logic that autonomously loads and evaluates goodness-of-fit metrics locally, effectively decoupling the semantic reasoning process from the computational burden of data storage.

Collectively, this structured integration of physical bounds, algorithmic protocols, and data schemas transforms the LLM from a generic probabilistic text generator into a domain-constrained reasoning engine. By strictly delineating the solution space and execution logic prior to code generation, the HydroCoder ensures that the resulting Python scripts are not only syntactically executable but also hydrogeologically consistent, providing a robust foundation for the subsequent runtime execution.

2.2.2. Executor: runtime control and noise filtering

Complementing the reasoning agent, the Executor functions as a robust runtime interface designed to bridge the semantic gap between the LLM's high-level logic and the rigorous physical engines. However, a critical bottleneck in coupling LLMs with external solvers is context window limitation: a single simulation run often generates thousands of lines of execution logs, which would instantly saturate the model's limited memory window.

To address this, we implemented a custom whitelist-based logging mechanism based on the SmartLogger class. This mechanism operates as a high-precision noise-reduction layer by intercepting the solver's raw stdout and stderr streams and performing real-time signal extraction. It systematically filters out low-level numerical noise, such as iteration steps and solver convergence details, while allowing only high-value keywords (e.g., "Final RMSE", "Traceback", and "Optimization Result") to reach the feedback stream. Simultaneously, the Executor follows a strict error-handling protocol. If execution fails due to syntax errors, library conflicts, numerical instabilities (e.g., matrix singularities), or runtime exceptions (e.g., I/O file locks), the full Python traceback is captured and formatted as a structured observation record.

By optimizing the signal-to-noise ratio of the feedback, this design ensures that the agent's reasoning capacity remains focused on high-level decision-making and trajectory adjustment rather than being overwhelmed by computational noise. This filtered feedback loop enables the framework to perform autonomous self-correction, diagnosing and patching script logic iteratively even within unstable simulation environments.

2.3. The iterative self-correction mechanism

One of the key features of the Hydro-Agent framework is its autonomous debugging capability. In complex scientific computing, rigid automation pipelines often fail due to minor interface discrepancies, library version conflicts, or dimensionality mismatches. To mitigate this inherent brittleness, our framework implements a closed-loop self-correction mechanism. Unlike traditional scripts that crash upon error, the system reinterprets code execution exceptions not as fatal failures, but as informative observations for policy refinement (Chen et al., 2023; Shinn et al., 2023).

This mechanism is exemplified by a representative implementation challenge encountered during the interface setup. In the initial calibration phase, the HydroCoder generated a Python script that attempted to interpret the return value of the physical simulation wrapper (*simulation_objective_calibration*) as a scalar float. However, the wrapper was configured to return a tuple containing comprehensive diagnostics: (series, mse, mae). Consequently, the Executor encountered a runtime `TypeError` when the optimizer attempted to minimize the tuple. Instead of aborting the workflow, the framework captured the full Python traceback and fed it back to the agent as a new observation. By analyzing the error message alongside the source code, the HydroCoder correctly identified the unpacking mismatch and autonomously rewrote the wrapper function to extract the third element (mae) as the scalar objective, effectively resolving the API misalignment without human

intervention.

We formalize this self-correction process as a sequential decision-making problem. Let S_t denote the calibration script generated at iteration t , the feedback observation O_t obtained from the environment is defined conditionally based on the execution status of S_t :

$$O_t = \begin{cases} M_t, & \text{if execution succeeds} \\ E_t, & \text{if execution fails} \end{cases} \quad (1)$$

where M_t represents the quantitative performance metrics (e.g., RMSE, MAE) returned upon successful simulation, and E_t denotes the structured error traceback (e.g., `SyntaxError`, `ValueError`) captured from the `stderr` stream.

The HydroCoder then functions as a probabilistic generator G , parameterized by the LLM, which produces the script for the next iteration S_{t+1} based on the static system constraints P and the dynamic interaction history H_t :

$$S_{t+1} = G(P, H_t) \quad (2)$$

where the history $H_t = \{(S_0, O_0), \dots, (S_t, O_t)\}$ accumulates the context of previous attempts and their consequences. This formulation allows the agent to effectively perform in-context learning, progressively refining the code logic until it satisfies both the syntactic requirements of the Python interpreter and the physical data structure of the groundwater model.

2.4. Implementation details and computational environment

The Hydro-Agent framework is instantiated using AutoGen (Wu et al., 2024), an open-source library that facilitates the orchestration of conversable agents via automated message passing. The computational backend operates within a Python 3.10 environment, integrating the `flopy` library for MODFLOW interface management and custom I/O parsers for TOUGHREACT execution. The foundational reasoning engine employed across all reasoning tasks is DeepSeek-V3 (Liu et al., 2024) (API deployment). This model was selected for its demonstrated proficiency on scientific coding benchmarks and its capability to process long-context hydrogeological datasets. To mitigate the inherent stochasticity of Large Language Models and ensure strict reproducibility, we enforced a deterministic inference protocol: the sampling temperature was fixed at 0, the random seed was set to 42, and the frequency penalty was disabled.

To balance computational cost with exploration depth, the framework operates under adaptive resource constraints tailored to the physics-based workflow and specific problem latency requirements. Unlike standard conversational agents limited by simple chat rounds, our Executor is governed by dynamically assigned optimization budgets defined by maximum iteration counts, function evaluation limits, and convergence thresholds that the HydroCoder tunes based on the specific case study. The context window is strictly managed within a 16 K token limit, utilizing a whitelist-based logging mechanism to prevent context window saturation during extended simulations. All numerical experiments were conducted on a standardized workstation equipped with an Intel Core i9-13,900 K processor and an NVIDIA GeForce RTX 4070 (12 GB VRAM) to ensure consistent runtime performance. Full details regarding system prompts and hyperparameter configurations are provided in Tables S1–S4 of the Supporting Information.

3. Experimental design

To systematically evaluate the versatility and robustness of the Hydro-Agent framework, we designed three progressively complex numerical experiments covering a spectrum of hydrogeological challenges. The validation hierarchy commences with Case I, a synthetic benchmark test focusing on identifying discrete geological structures within a

heterogeneous steady-state flow field. This is followed by Case II, which provides a rigorous assessment of one-dimensional reactive transport to examine the agent's ability to resolve non-linear competitive cation exchange kinetics under substantial stochastic noise. The evaluation concludes with Case III, a real-world, field-scale application to the Aquia Aquifer, challenging the agent to calibrate a high-dimensional multi-component system governed by complex thermodynamic equilibria. Collectively, this hierarchical design assesses the agent's autonomous reasoning across diverse physics engines (MODFLOW, TOUGHREACT) and its adaptability to varying degrees of spatial heterogeneity and parametric uncertainty. In addition, a baseline comparison was conducted by directly coupling the physics solvers with the optimizer without LLM mediation, using identical optimization settings and hyperparameter configurations, to assess whether the agent-driven workflow alters inversion outcomes.

3.1. Case I: identification of hydraulic heterogeneity in steady-state flow

This case serves as a benchmark for structural parameter identification, validating the framework's ability to resolve complex geological zonation solely from hydraulic head observations. We constructed a synthetic truth model representing a tectonically controlled alluvial plain (2000×2000 m), simulated using the MODFLOW-2005 engine via the flopy interface. The domain is discretized into a single layer of 40×40 finite difference cells ($\Delta x = \Delta y = 50$ m). Dirichlet boundary conditions are imposed to establish a regional hydraulic gradient, with constant heads of $H = 45$ m along the west boundary and $H = 25$ m along the east boundary. The steady-state flow equation is solved using the Pre-conditioned Conjugate Gradient (PCG) solver with a strict head change convergence criterion of 10^{-5} m.

The hydraulic conductivity (K) field is parameterized by seven distinct geological zones as delineated in Table 1 and Fig. 2. These zones create a challenging, non-convex optimization landscape characterized by sharp property contrasts. Specifically, a high-permeability, sinusoidal paleochannel defined by the function $y = 20 + 8\sin(x/6)$ acts as a preferential flow path; this feature is physically segmented into upstream gravel and downstream sand facies to test the agent's ability to resolve longitudinal heterogeneity. Intersecting this depositional system is a diagonal conductive fault zone ($y = -x + 35$) acting as a structural

Table 1

Parameterization of the synthetic heterogeneous aquifer model (Case I), including geological descriptions, true values, and prior search ranges.

Parameter	Geological zone description	True value (K , m/d)	Prior search range (m/d)
K_{North}	Northern floodplain	2.0	[0.1, 100.0]
K_{South}	Southern floodplain	5.0	[0.1, 100.0]
$K_{Paleo,up}$	Paleochannel upstream	40.0	[0.1, 100.0]
$K_{Paleo,down}$	Paleochannel downstream	25.0	[0.1, 100.0]
K_{Clay}	Isolated clay lens	0.01	[0.001, 1.0]
K_{Sand}	Isolated sand lens	35.0	[1.0, 100.0]
K_{Fault}	Conductive fault zone	80.0	[1.0, 100.0]

discontinuity. Additionally, the domain includes isolated local anomalies: a low-permeability clay lens acting as a hydraulic barrier and a high-permeability sand lens functioning as a local reservoir. To enhance parameter identifiability and mitigate equifinality between the fault and the channel, the system is hydrodynamically stressed by two pumping wells located at $(x = 750$ m, $y = 1000$ m) and $(x = 1500$ m, $y = 500$ m) with extraction rates of $Q_1 = 2000$ m³/d and $Q_2 = 500$ m³/d, respectively. These stresses induce convergent flow patterns that actively interrogate the subsurface and accentuate the hydraulic signatures of the low-permeability zones. The inverse problem tasks the agent with reconstructing the true K values of all seven zones using noisy head observations ($\sigma = 0.05$ m) derived from nine monitoring wells. These observation points were strategically distributed to capture critical hydraulic gradients across the facies boundaries and within the cone of depression.

3.2. Case II: 1D reactive transport under stochastic uncertainty

The second case evaluates the framework's precision and noise tolerance in capturing non-linear chemical kinetics. Using the TOUGHREACT engine, we simulate a one-dimensional porous column ($L = 0.08$ m) discretized into 20 grid blocks. The medium is homogeneous, characterized by a porosity of 0.3 and a bulk density of 2650 kg/m³. The system is initially saturated with a mixed background solution (NaNO₃ and KNO₃). A tracer solution containing CaCl₂ (0.6 mmol/L) is injected

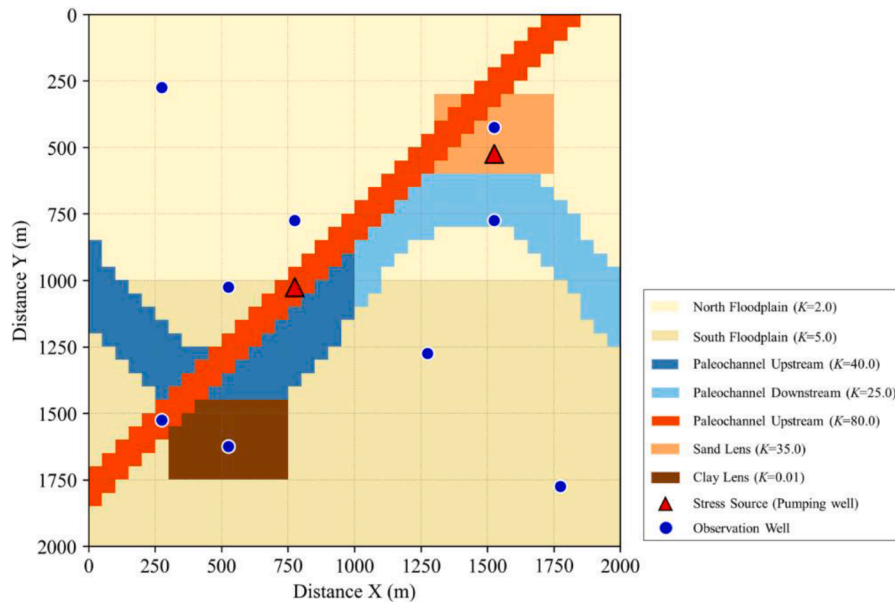


Fig. 2. Conceptual model and true K field for Case I.

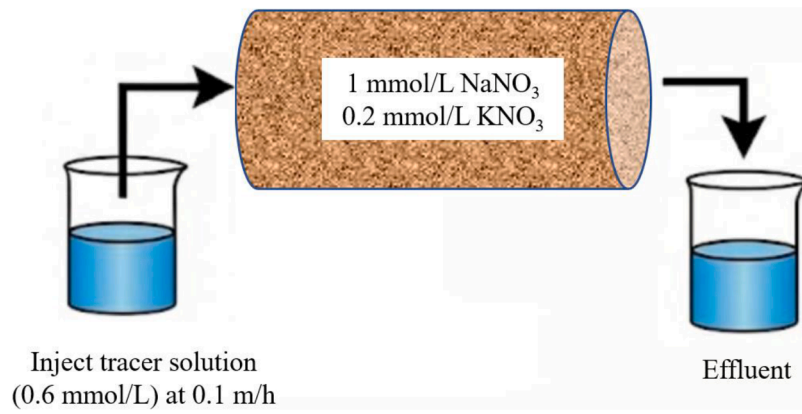


Fig. 3. Schematic representation of the 1D reactive transport column experiment (Case II).

Table 2

Hydrogeochemical parameters for the 1D column transport experiment (Case II). True values represent the synthetic ground truth, and prior ranges define the search space.

Parameter Type	Symbol	True value	Prior interval	Unit
Selectivity coefficient	$K_{Na/K}$	0.1995	[0.1, 0.3]	-
	$K_{Na/Ca}$	0.3981	[0.25, 0.42]	-
Initial concentration	C_{Na}	1.0×10^{-3}	$[8.0 \times 10^{-4}, 8.0 \times 10^{-3}]$	mol/L
	C_K	2.0×10^{-4}	$[1.0 \times 10^{-4}, 1.0 \times 10^{-3}]$	mol/L
Boundary concentration	$C_{Ca,b}$	6.0×10^{-4}	$[3.0 \times 10^{-4}, 8.0 \times 10^{-4}]$	mol/L
	$C_{Cl,b}$	1.2×10^{-3}	$[5.0 \times 10^{-4}, 5.0 \times 10^{-3}]$	mol/L
Exchange Capacity	CEC	0.01779	[0.005, 0.05]	meq/100 g

from the inlet boundary at a constant Darcy velocity of 0.1 m/h (Fig. 3). The geochemical system is governed by instantaneous cation exchange reactions utilizing the Gaines-Thomas selectivity convention. The reaction network is coupled with the Extended Debye-Hückel model for aqueous species activity corrections, primarily involving competition for surface sites (X): $Na^+ + 0.5Ca-X_2 = 0.5Ca^{2+} + Na-X$ and $K^+ + Na-X = K-X + Na^+$.

The inversion campaign challenges the Hydro-Agent to simultaneously estimate seven hydrogeochemical parameters (Table 2): two selectivity coefficients ($K_{Na/K}$, $K_{Na/Ca}$), four initial and boundary concentrations, and the CEC. Crucially, to replicate realistic laboratory measurement errors, we introduced three levels of multiplicative Gaussian noise ($\sigma = 1\%$, 5% , and 10%) into the synthetic breakthrough curves of all four ionic species (Cl^- , Na^+ , K^+ , Ca^{2+}) to rigorously evaluate the framework's robustness under varying measurement uncertainties. The agent is explicitly prompted to search for the global optimum under these heteroscedastic noises, requiring the autonomous configuration of robust objective functions that prevent overfitting to high-concentration outliers.

3.3. Case III: field-scale multicomponent calibration

The final case applies the framework to a classic hydrogeological benchmark: the Aquia Aquifer in Maryland, USA. This scenario represents a substantial escalation in complexity, addressing a high-dimensional, stiff reactive transport problem widely studied in the literature (Chen et al., 2021; Dai and Samper, 2006). The conceptual model is formulated as a 1D reactive transport simulation along a streamline, covering a 96.56 km flow path from the recharge zone to the discharge area (Fig. 4). The domain is discretized into 40 grid blocks and simulated over a geological timeframe of 10^5 years to reach a quasi-steady state. The dominant hydrogeochemical process is the natural freshening of the aquifer, where recharging fresh water progressively displaces the connate brackish water. As groundwater flows downstream, the water quality evolves from a calcium-magnesium-bicarbonate ($Ca-Mg-HCO_3$) facies to a sodium-bicarbonate ($Na-HCO_3$) facies. This evolution is driven by a complex, coupled reaction network including cation exchange on glauconite, equilibrium dissolution/precipitation of mineral phases (calcite, dolomite), and, critically, proton exchange on clay minerals which buffers the pH evolution (Appelo, 1994).

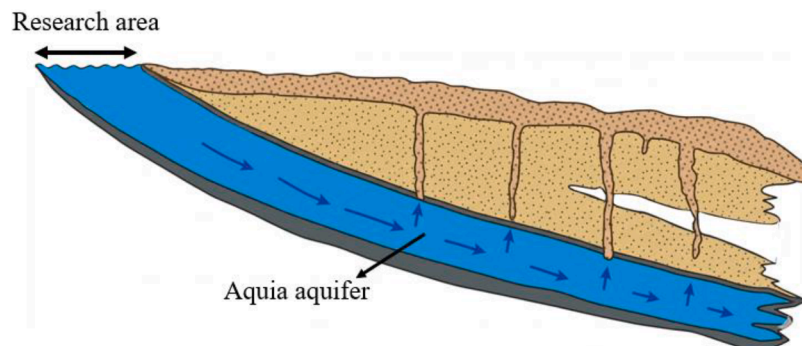


Fig. 4. Hydrogeological profile of the Aquia Aquifer (Case III). (Revised from Dai and Samper (2006) and Chen et al. (2021)).

Table 3

Prior constraints for the 18 parameters of the field-scale Aquia Aquifer model (Case III). Prior intervals are derived from regional hydrogeological studies (Dai and Samper, 2004; Chen et al., 2022).

Category	Parameter	Symbol	Prior search interval	Unit
Physical	Permeability	k	$[1.0 \times 10^{-12}, 9.9 \times 10^{-12}]$	m^2
Cation exchange	Selectivity coefficient	$K_{Na/K}$	$[0.007, 0.5]$	-
		$K_{Na/Ca}$	$[0.1, 0.9]$	-
		$K_{Na/Mg}$	$[0.1, 0.9]$	-
		$K_{Na/H}$	$[1.0 \times 10^{-6}, 8.0 \times 10^{-6}]$	-
Initial condition	Exchange capacity	CEC	$[1.0, 6.0]$	meq/100 g
	pH	pH_0	$[6.5, 9.0]$	-
	Concentrations	C_{Ca}	$[0.001, 0.009]$	mol/L
		C_{Mg}	$[0.007, 0.04]$	mol/L
		C_{Na}	$[0.05, 0.1]$	mol/L
		C_K	$[0.001, 0.004]$	mol/L
		C_{HCO3}	$[0.008, 0.04]$	mol/L
Boundary condition	pH	pH_B	$[6.5, 9.0]$	-
	Concentrations	$C_{Ca,b}$	$[0.0008, 0.004]$	mol/L
		$C_{Mg,b}$	$[8.0 \times 10^{-7}, 4.0 \times 10^{-6}]$	mol/L
		$C_{Na,b}$	$[7.0 \times 10^{-5}, 4.0 \times 10^{-4}]$	mol/L
		$C_{K,b}$	$[2.0 \times 10^{-5}, 9.0 \times 10^{-5}]$	mol/L
		$C_{HCO3,b}$	$[0.001, 0.007]$	mol/L

Unlike the preceding synthetic experiments, this field-scale application imposes a high-dimensional and mathematically stiff optimization landscape characterized by severe scale disparities. As detailed in Table 3, the Hydro-Agent is tasked with the simultaneous calibration of 18 coupled parameters, including the intrinsic permeability (k), four distinct cation exchange selectivity coefficients ($K_{Na/K}$, $K_{Na/Ca}$, $K_{Na/Mg}$, $K_{Na/H}$), the CEC, and a comprehensive set of 12 initial and boundary condition parameters governing the major ion concentrations (Na^+ , K^+ , Ca^{2+} , Mg^{2+} , HCO_3^-) and pH. This vast dynamic range, spanning from major ions (Na^+ , Ca^{2+} at $\sim 10^{-3}$ mol/L) to trace protons (H^+ at $\sim 10^{-8}$ mol/L), creates a critical stress test due to severe parameter correlations and magnitude dominance. This effectively challenges the agent's ability to autonomously navigate rugged, ill-conditioned solution surfaces and formulate weighted objective functions.

4. Results and discussion

To rigorously validate the performance and adaptability of the Hydro-Agent framework, we present a systematic evaluation across the three distinct hydrogeological scenarios defined in Section 3. Beyond routine parameter estimation, the agent functions as an autonomous strategist, dynamically adjusting optimization algorithms and hyperparameters based on numerical feedback. These adaptive configurations

and their underlying decision logic are detailed in Text S4 and Table S5. The analysis concludes with a critical assessment of the framework's efficiency, interpretability, and the implications of human-on-the-loop supervision.

4.1. Performance in heterogeneity reconstruction (Case I)

4.1.1. Optimization dynamics and convergence efficiency

The evolution of the objective function reveals the framework's search efficiency in a high-dimensional landscape (Fig. 5). The RMSE trajectory exhibits a distinct two-stage convergence behavior, driven by the agent's autonomous strategy hybridization. The HydroCoder initiated the process with a DE global search to navigate the non-convex landscape, and subsequently handed off the optimization to a gradient-based solver (L-BFGS-B) for high-precision refinement. The process begins with an exploratory phase (iterations 0–1200), where the agent utilizes the stochastic mutation of DE to avoid local optima arising from discrete geological zonation. This transition from global exploration to local exploitation allowed the population to converge rapidly towards the global basin of attraction, performing progressive parameter refinement. After approximately 3500 model evaluations, the system achieved stable convergence, with the RMSE decreasing from an initial value of 0.625 m to 0.0342 m.

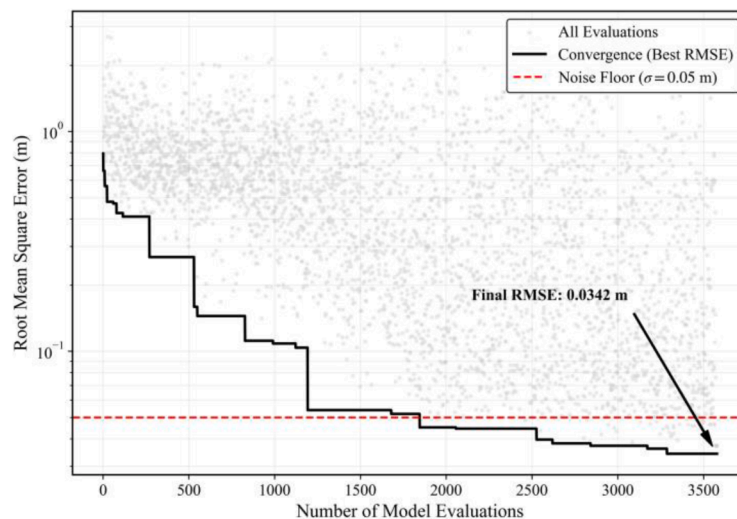


Fig. 5. Convergence history of the Hydro-Agent framework for Case I.

Table 4
Comparison of ground truth and AI-inverted parameters for Case I.

Parameter	Geological feature	True value (m/d)	Inverted value (m/d)	Qualitative assessment
K_{North}	Northern floodplain	2.0	2.95	Accurate retrieval
K_{South}	Southern floodplain	5.0	5.74	Accurate retrieval
$K_{Paleo,up}$	Paleochannel upstream	40.0	28.43	High- K trend captured
$K_{Paleo,down}$	Paleochannel downstream	25.0	13.38	Moderate- K trend captured
K_{Clay}	Isolated clay lens	0.01	0.71	Low- K barrier identified
K_{Sand}	Isolated sand lens	35.0	72.76	High- K anomaly resolved
K_{Fault}	Conductive fault zone	80.0	91.46	High- K conduit identified

A key finding is that the final converged RMSE (0.0342 m) consistently remains below the theoretical noise floor defined by the injected Gaussian standard deviation (≈ 0.05 m). This convergence below the nominal noise floor suggests that the Hydro-Agent effectively decoupled the deterministic hydraulic signal from the superimposed stochastic noise. This capability is attributed to the structural regularization imposed by the groundwater flow equation itself; the diffusive nature of the governing partial differential equations as an intrinsic physical low-pass filter, naturally suppressing high-frequency noise artifacts that are incompatible with the continuum mechanics of flow. By leveraging this property, the framework retrieved a parameter set that reflects the underlying aquifer mechanics rather than observational artifacts.

4.1.2. Structural contrasts and parameter sensitivity

Table 4 presents a quantitative comparison between the ground-truth values and the AI-inverted parameters. Relative errors were calculated as $|\theta_{true} - \theta_{inv}|/\theta_{true}$, and they vary substantially across different geological units. Major flow-controlling structures such as the conductive fault (K_{Fault} : 91.46 vs. 80.00 m/d) and regional floodplain matrices (K_{North} : 2.95 vs. 2.00 m/d; K_{South} : 5.74 vs. 5.00 m/d) exhibit moderate deviations, whereas isolated anomalies (e.g., K_{Clay} and K_{Sand}) show significantly larger discrepancies. Notably, K_{South} and K_{Fault} are within $\sim 15\%$ range, while other parameters exceed this threshold.

To clarify this variability, we conducted a local sensitivity analysis by perturbing each parameter by $\pm 10\%$ around its inverted value and evaluating the corresponding change in hydraulic head RMSE

(Supplementary Figure S1). The results indicate that K_{Fault} and the regional floodplain conductivities exert the strongest control on the objective function, whereas K_{Clay} and K_{Sand} exhibit minimal sensitivity. This indicates that the larger deviations observed in low-permeability or localized features correspond to weakly identifiable parameters under the available head observation network and can be compensated by other parameters with limited impact on head misfit. Despite substantial absolute deviation for K_{Clay} (0.01 vs. 0.71 m/d), the inversion preserves its functional role as a hydraulic barrier. This reflects classical equifinality and parameter compensation behavior: once conductivity falls below a critical contrast threshold, further reduction produces negligible change in the regional head distribution. Therefore, the inversion should be interpreted as structurally consistent rather than uniquely determined at the parameter level.

Crucially, the execution logs indicate that the Hydro-Agent's reasoning extends beyond numerical optimization to include conceptual validation. As recorded in the execution logs synthesized in real-time, the agent explicitly quantified a fault dominance ratio of 3.1:1 and a south/north regional contrast of 2.67:1. Using these derived metrics, it correctly concluded that the fault zone controls $>80\%$ of the system's total flux. This ability to perform qualitative reasoning demonstrates that the large language model does not merely treat parameters as abstract scalar variables but actively constructs a mechanistic understanding of the aquifer's hydrodynamic behavior—a capability that has traditionally remained the exclusive domain of human experts.

4.1.3. Spatial residual and structural fidelity

The spatial fidelity of the inversion is rigorously diagnosed through the hydraulic head residual map (Fig. 6). A visual comparative analysis reveals that the AI-inverted head field (Fig. 6b) successfully replicates the complex flow refraction and gradient steepening (Fig. 6a) induced by the curvilinear paleochannel and the diagonal fault zone. As illustrated in the residual map (Fig. 6c), the hydraulic head differences ($H_{sim} - H_{obs}$) exhibit a near-zero mean distribution across the vast majority of the aquifer, indicating that the dominant hydraulic structures are effectively captured.

However, localized high-magnitude residuals are observed in regions corresponding to the isolated sand lens and clay lens units. These zones also exhibit the largest deviations in recovered hydraulic conductivity (Table 4), suggesting that the observed head discrepancies primarily stem from parameter identifiability challenges rather than numerical discretization artifacts. In steady-state potential flow systems, once the hydraulic conductivity of a localized unit reaches a high contrast relative to the surrounding matrix, the hydraulic head field becomes inherently insensitive to further parameter perturbations. Consequently,

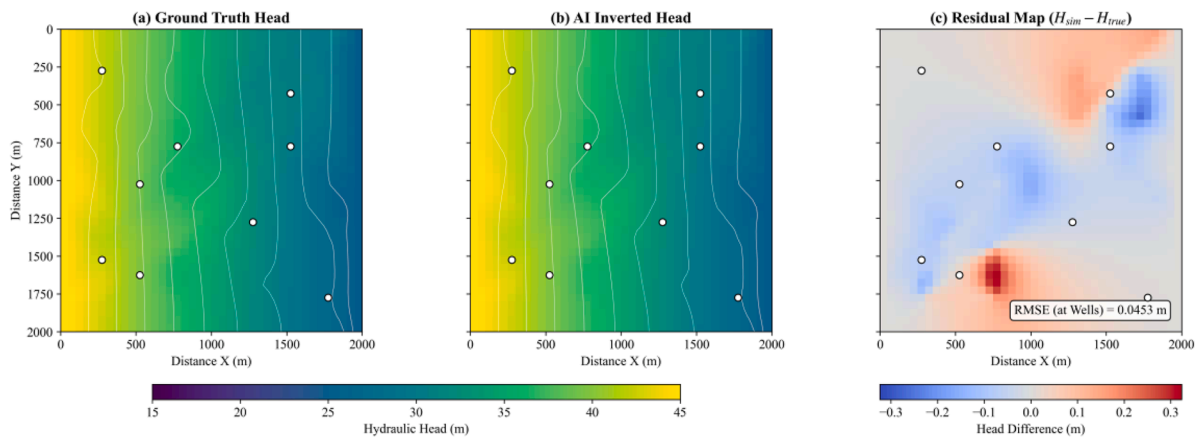


Fig. 6. Spatial error analysis. (a) The ground truth hydraulic head field. (b) The simulated head field using AI-inverted parameters. (c) The residual map ($H_{sim} - H_{obs}$).

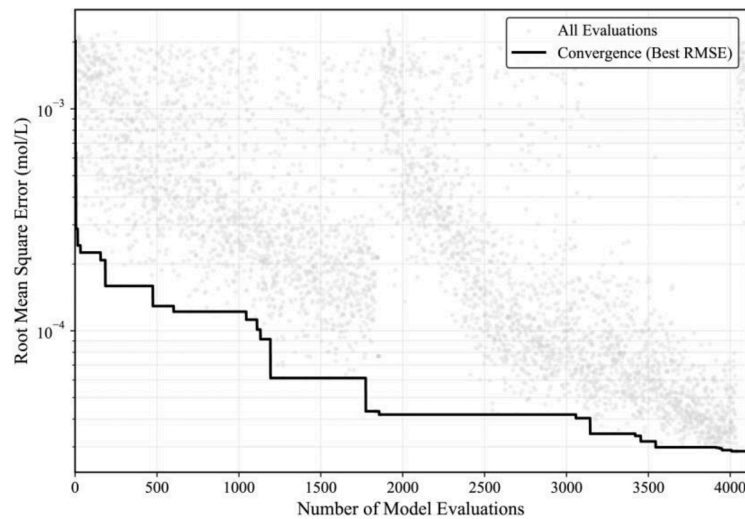


Fig. 7. Representative convergence history of the Hydro-Agent framework for Case II (demonstrated under the 5% multiplicative noise scenario).

large relative deviations in conductivity within such low-sensitivity zones may induce only limited fluctuations in hydraulic head. Therefore, while certain localized units exhibit substantial parameter deviations, the inversion remains structurally consistent in reproducing the essential hydraulic contrasts governing regional flow behavior. This residual pattern reflects classical non-uniqueness in groundwater inversion rather than limitations of the numerical framework.

4.2. Robustness in geochemical parameter retrieval (Case II)

4.2.1. Resilience to observational noise

The autonomous calibration exhibited a robust convergence trajectory (Fig. 7), reducing the Mean Squared Error (MSE) by over two orders of magnitude under varying noise levels. For the 1%, 5%, and 10% noise scenarios, the final stabilized RMSE values converged to 1.296×10^{-5} , 2.865×10^{-5} , and 5.630×10^{-5} , respectively. These results represent

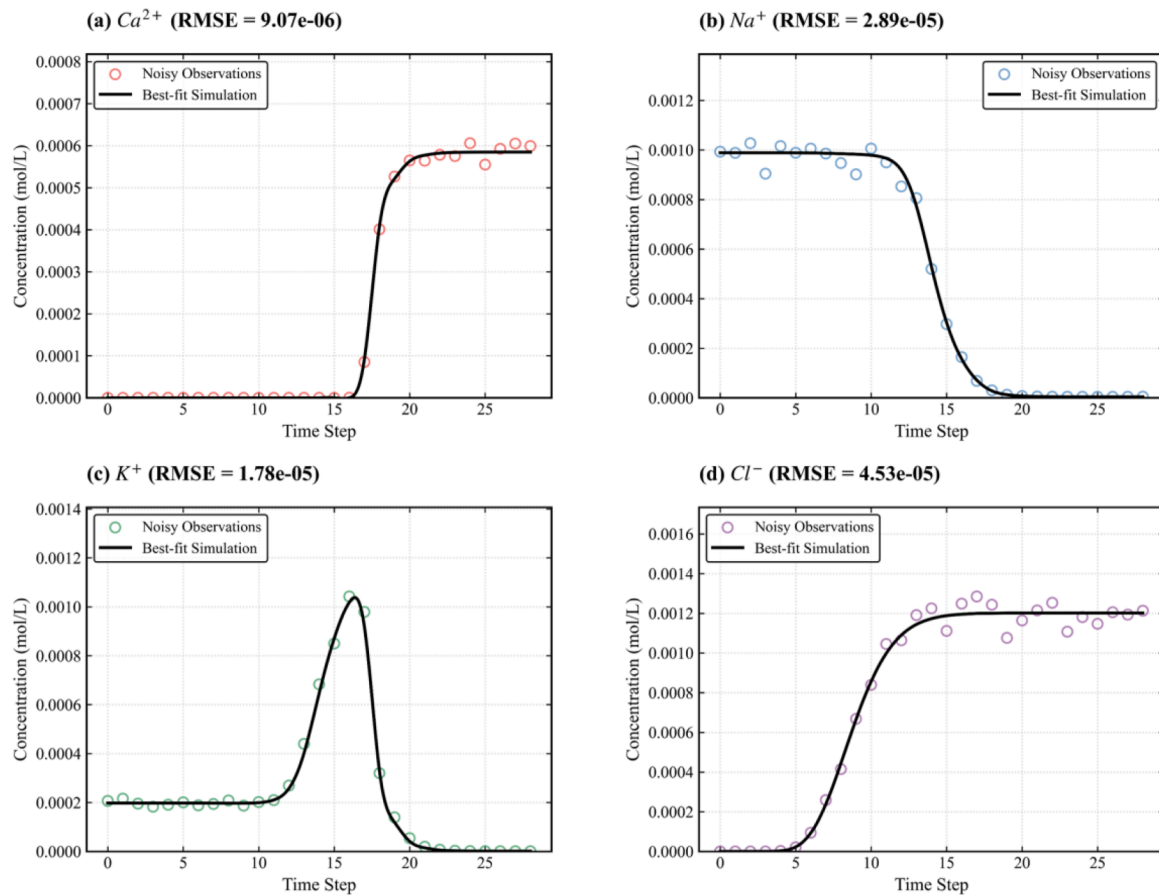


Fig. 8. Comparison of noisy synthetic observations and AI-calibrated breakthrough curves for Case II (representative results under 5% noise level). The subplots (a–d) represent the temporal evolution of Ca^{2+} , Na^{+} , K^{+} , and Cl^{-} concentrations at the column outlet.

sub-noise precision levels significantly lower than the nominal amplitude of the stochastic perturbations, demonstrating that the residual error scales linearly with the noise magnitude rather than overfitting stochastic artifacts.

In the context of reactive transport, this result confirms that the Hydro-Agent functioned effectively as a physics-informed regularizer. Unlike purely data-driven surrogates that risk overfitting stochastic artifacts, the agent is constrained by the rigid thermodynamic principles of the Gaines-Thomas mass action laws and elemental mass conservation. These governing equations act as a strong structural filter, preventing the optimizer from fitting high-frequency noise that violates chemical equilibrium. Consequently, the framework successfully decoupled the latent deterministic geochemical signal from the heteroscedastic observational noise, recovering the intrinsic kinetic parameters without manual regularization.

The efficiency of the convergence, characterized by a rapid initial descent followed by a stable asymptotic approach to the global minimum, validates the efficacy of the multi-stage optimization policy autonomously configured by the HydroCoder. To break through the final convergence bottleneck, the agent transitioned from the stochastic DE algorithm to a Truncated Newton (TNC) solver, tightening the tolerance from 0.01 to 1×10^{-12} to ensure a rigorous local minimum. Beyond algorithmic robustness, the findings highlight the framework's potential for operational deployment. Unlike standard optimization algorithms that imply manual post-processing, the HydroCoder autonomously encapsulated the calibration logic into a production-ready workflow. The agent implemented strict exit-code protocols (e.g., executing `sys.exit(0)` for "SYSTEM_OK" only if $\text{RMSE} < 1.2 \times 10^{-4}$) and generated automated crontab scheduling instructions for daily execution. This behavior illustrates that the agent's policy encompasses robust software engineering practices, ensuring that the inverted model can be seamlessly integrated into automated early-warning systems for contaminant transport without requiring secondary code refactoring.

4.2.2. Reproduction of chromatographic separation dynamics

The geochemical fidelity of the inversion is substantively evidenced by the precise alignment between the noisy observations and the AI-calibrated simulation results (Fig. 8). The framework successfully captured the complex, transient behaviors of all four ionic species (Ca^{2+} , Na^+ , K^+ , and Cl^-), effectively distinguishing between the conservative transport of Cl^- and the reactive retardation of cations. Specifically, the agent replicated the selectivity-driven displacement chromatography, where the invading divalent Ca^{2+} front progressively displaces the monovalent Na^+ and K^+ from the mineral exchange sites. This dynamic is governed by the valence effect inherent in the Gaines-Thomas convention, which favors the adsorption of higher-charge species over lower-charge ones as the ionic strength evolves.

Most notably, the agent accurately resolved the characteristic elution peaks (concentration overshoots) observed for Na^+ and K^+ (Fig. 8b, 8c). These temporary peaks arise from the competitive exchange process and

are highly sensitive to the selectivity coefficients ($K_{\text{Na/K}}$, $K_{\text{Na/Ca}}$) and the CEC. The successful reconstruction of these non-linear features demonstrates that the Hydro-Agent did not merely perform a mathematical curve-fitting but successfully parameterized the underlying laws of mass action, effectively filtering the stochastic noise to reveal the true thermodynamic response of the system.

As indicated by the statistical metrics, the fitting precision remains high across all concentration scales. This consistent accuracy confirms that the agent successfully decoupled the reactive transport mechanism from the multiplicative stochastic interference, ensuring that the retrieved parameters adhere to the stoichiometric constraints of the conceptual model regardless of the uncertainty magnitude.

4.2.3. Geochemical consistency verification

A comparative analysis between the inverted parameters and their respective ground truths across three noise levels (1%, 5%, and 10%) reveals the framework's high-fidelity retrieval capability (Table 5). While the final RMSE scales proportionally with the noise magnitude, the estimated geochemical parameters remain remarkably stable, indicating that the framework successfully avoids overfitting stochastic artifacts. In multi-component reactive transport, selectivity coefficients and exchange capacity are fundamentally coupled, as both dictate the retardation and magnitude of cation breakthrough. Despite this numerical coupling, the Hydro-Agent consistently preserved the relative selectivity order ($K_{\text{Na/Ca}} > K_{\text{Na/K}}$). This identifiability is facilitated by the framework's strict adherence to mass action and conservation laws. By requiring every iteration to honor these thermodynamic principles, the embedded physics engine serves as a structural regularizer, restricting the search space to physically plausible manifolds and enabling the agent to resolve deterministic geochemical signals through significant heteroscedastic interference. Nevertheless, the slight deviations observed in specific parameters should be viewed as a joint manifestation of structural equifinality and its synergistic interaction with observational noise, which inherently limits the absolute precision of the inversion.

Furthermore, the stability of the retrieved boundary and initial concentrations across all uncertainty levels demonstrates an effective decoupling of transport-related boundary conditions from reaction-related kinetic constants. Even under the 10% noise scenario, a magnitude often encountered in real-world sensor data, the maximum relative error for key parameters remained within an acceptable range. This suggests that the autonomously configured optimization strategy successfully parameterized the mechanistic response of the aquifer rather than performing a mere numerical minimization of residuals. This sustained precision substantively addresses the challenge of geochemical equifinality, where breakthrough behaviors are often subject to trade-offs between site density and relative affinities. Such robustness provides a reliable pathway for operational data assimilation, ensuring that inverted models remain physically consistent and suitable for integration into automated contaminant monitoring and early-warning systems where measurement uncertainty is inevitable.

4.3. Scalability to field-scale reactive transport (Case III)

4.3.1. Adaptive optimization strategies for high-dimensional spaces

A critical hurdle encountered during the calibration of the high-dimensional Aquia system was the "magnitude dominance" effect. Initial attempts utilizing a generic minimization goal revealed that the objective function became disproportionately sensitive to high-concentration species, thereby neglecting trace components essential for geochemical consistency. To rectify this, the optimization strategy was adapted under physics-guided supervision. Instead of relying on the agent's initial raw RMSE formulation, we introduced a hybrid log-transformed weighted objective function into the logic loop. This human-on-the-loop intervention effectively normalized residuals across varying concentration scales, enforcing a balanced optimization weight

Table 5

Comparison of ground truth and AI-inverted geochemical parameters (Case II).

Parameter category	Symbol	True value	1% noise	5% noise	10% noise
Selectivity coefficient	$K_{\text{Na/K}}$	0.1995	0.1997	0.2057	0.2224
	$K_{\text{Na/Ca}}$	0.3981	0.3842	0.2851	0.2955
Initial concentration	C_{Na}	1.0×10^{-3}	9.99×10^{-4}	9.89×10^{-4}	9.62×10^{-4}
	C_{K}	2.0×10^{-4}	1.94×10^{-4}	1.98×10^{-4}	2.14×10^{-4}
	$C_{\text{Ca,b}}$	6.0×10^{-4}	5.80×10^{-4}	5.87×10^{-4}	5.84×10^{-4}
	$C_{\text{Cl,b}}$	1.2×10^{-3}	1.20×10^{-3}	1.21×10^{-3}	1.22×10^{-3}
Exchange capacity	CEC	0.01779	0.01707	0.01727	0.01733

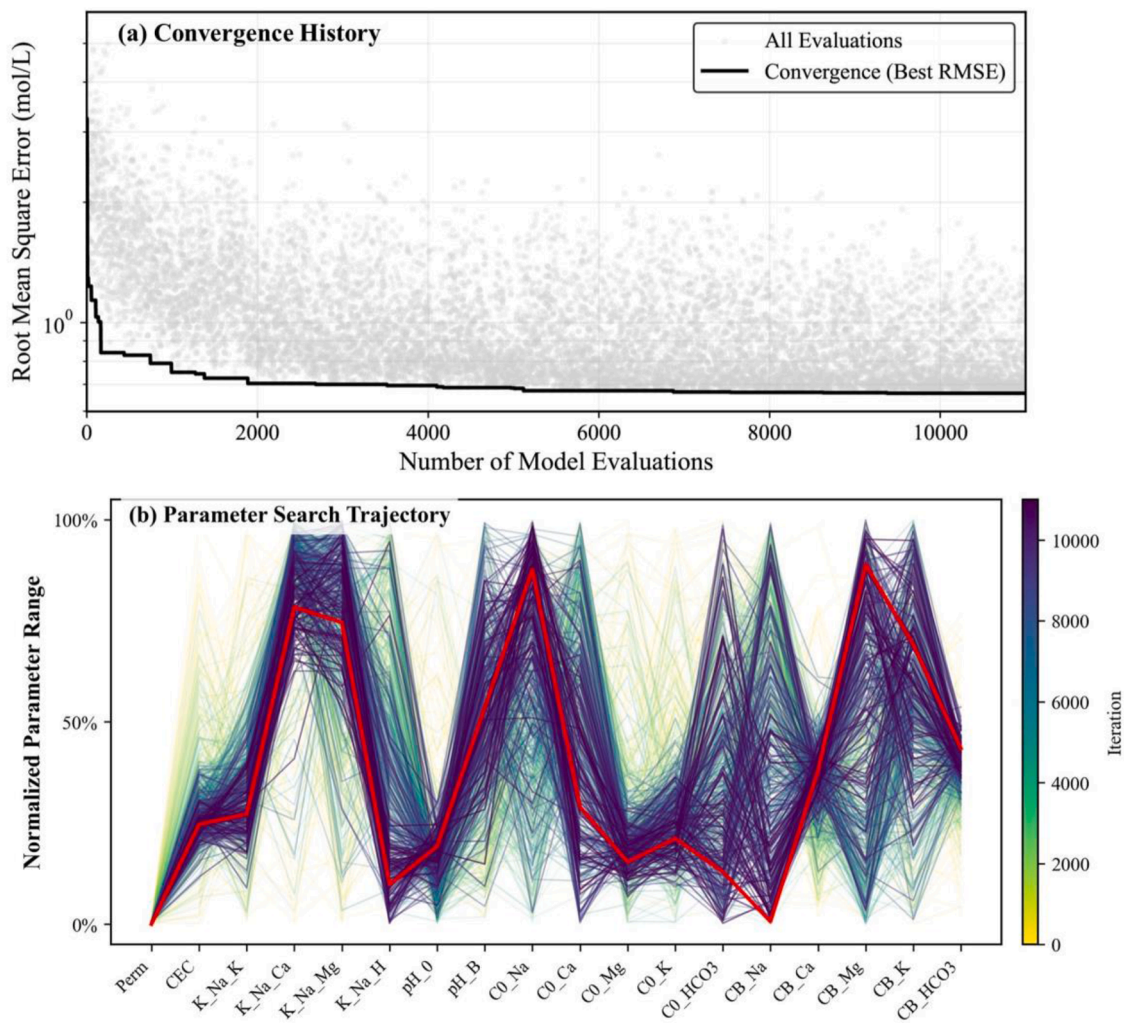


Fig. 9. Optimization dynamics in high-dimensional parameter space. (a) Convergence history of the Hydro-Agent framework for Case III. (b) Parallel coordinate plot showing the normalized trajectories of the 18 hydrogeochemical parameters. The lines color-coded by iteration number (light to dark) demonstrate how the agent successfully narrowed the search from random initializations to a stable, physically consistent solution set.

between the dominant freshening front and the subtle pH buffering mechanisms.

With this physically consistent objective function defined, the HydroCoder agent effectively navigated the rugged optimization landscape. The efficacy of this collaborative strategy is visualized in Fig. 9, which tracks the convergence trajectory within the 18-dimensional parameter space. Fig. 9a demonstrates a rapid descent in the global objective function, identifying the primary basin of attraction within the first 1500 evaluations. Subsequently, the search transitions into an asymptotic refinement phase. The parallel coordinate plot (Fig. 9b) further reveals how the ensemble of candidate parameters, initially scattered across the full prior range, progressively collapses into a coherent, physically consistent solution set. This convergence behavior indicates that the agent-engineered optimization pipeline successfully identifies a stable basin. The HydroCoder recognized that random sampling is statistically inefficient in an 18-dimensional space; it thus implemented a heuristic warm-start (init=init_pop) and a custom persistence layer to ensure reliability against hardware interruptions during the intensive 10^5 -year simulations.

4.3.2. Comparative performance and hydrogeochemical interpretation

The numerical model of the Aquia aquifer has served as a critical benchmark for reactive transport inversion, having been previously calibrated by Dai and Samper (2006) using the gradient-based Gauss-Newton-Levenberg-Marquardt (GNLM) method, and more recently by

Chen et al. (2021) using a deep learning-based tandem neural network (TNNA-AUS). In this study, our Hydro-Agent was applied to the same 18-parameter estimation problem, achieving a comparable convergence level in curve fitting while demonstrating distinct advantages in automation and interpretability.

The physical fidelity of the Hydro-Agent's inversion is evidenced by the strong agreement between simulated and observed concentration profiles along the 60 km flow path (Fig. 10). The framework accurately reproduced the canonical chromatographic separation characteristic of the Aquia aquifer's natural freshening process. As illustrated in the profiles, the model captures the sequential displacement of native Na^+ by recharging Ca^{2+} and Mg^{2+} . This is quantitatively supported by the sharp decline of divalent cations and the concomitant rise of sodium, confirming the accurate identification of cation selectivity coefficients ($K_{Na/Ca}$, $K_{Na/Mg}$) and the CEC. Notably, the framework successfully captured the non-monotonic evolution of Mg^{2+} and K^+ , which peak at approximately 25 km and 35 km respectively (Fig. 10c, 10e). Regarding pH evolution (Fig. 10a), the model captures the rising trend from approximately 7.8 in the recharge zone to over 8.5 downstream. Accurately simulating this pH front is particularly challenging given that proton concentrations are orders of magnitude lower than the major ions. To address this scale disparity, we employed an inverse standard-deviation weighted RMS as the objective function. The precise fit validates the effectiveness of this weighting strategy in preserving sensitivity for proton exchange coefficients ($K_{Na/H}$) and carbonate species

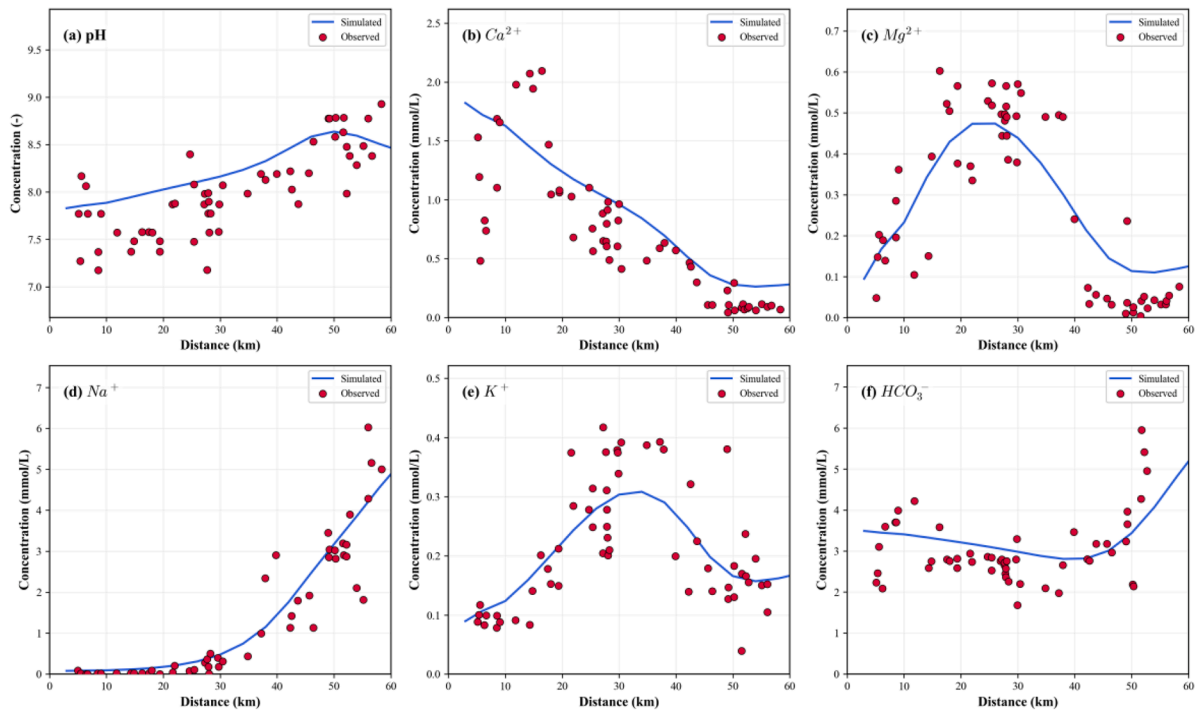


Fig. 10. Calibrated reactive transport profiles along the Aquia aquifer flow path. Comparison of observed (red dots) and AI-simulated (blue lines) concentrations for (a) pH, (b) Ca^{2+} , (c) Mg^{2+} , (d) Na^+ , (e) K^+ , and (f) HCO_3^- . (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

(HCO_3^-) is not overwhelmed by dominant ions.

Compared to the gradient-based approach (Dai and Samper, 2006), the Hydro-Agent demonstrates superior algorithmic robustness. Gradient-based methods like GNLM are notoriously sensitive to initial guess values and are prone to getting trapped in local minima within the rugged, non-convex error landscapes characteristic of multi-component reactive transport. Furthermore, GNLM requires the complex calculation of a Jacobian matrix, which is computationally fragile in stiff systems. In contrast, the Hydro-Agent functions as an autonomous reasoning engine; it bypasses the need for manual Jacobian setup by utilizing its HydroCoder to intelligently select, configure and execute global optimization strategies based on the problem's complexity, a task that typically requires expert-level intuition. This allows the framework to navigate high-dimensional parameter spaces and find physically consistent solutions without the extensive manual trial-and-error required in legacy methods.

When compared to the deep learning method (Chen et al., 2021), the

Hydro-Agent offers a fundamentally different paradigm in terms of interpretability and physical consistency. While the TNNA-AUS method leverages neural networks for computational speed, it functions as a statistical surrogate that is susceptible to approximation errors and often yields solutions that violate fundamental chemical constraints (e.g., charge balance and mass action laws), especially when requires a substantial initial dataset to train the forward network. As noted by Chen et al. (2021), such surrogate-based inversions can be biased if the initial training sample size is insufficient. The Hydro-Agent, however, operates on a “code-as-policy” principle. Instead of replacing physics with black-box neural weights, it directly orchestrates the actual physics-based simulator. Every iteration is grounded in rigorous thermodynamic laws, strictly enforcing charge neutrality and multicomponent equilibrium constraints. This inherently eliminates the approximation bias and the need for high-density pre-training datasets. By iteratively debugging its own code based on simulation feedback, the agent emulates the logic of an expert hydrogeologist, ensuring that the

Table 6

Comparison of prior search ranges and AI-inverted parameters for the Aquia aquifer model (Case III).

Parameter description	Symbol	Prior range	Inverted value	Unit
Intrinsic permeability	k	$[1.0, 9.9] \times 10^{-12}$	1.00×10^{-12}	m^2
Cation exchange capacity	CEC	$[1.0, 6.0]$	2.2332	meq/100 g
Selectivity coefficient	$K_{\text{Na}/\text{K}}$	$[0.007, 0.5]$	0.1413	-
	$K_{\text{Na}/\text{Ca}}$	$[0.1, 0.9]$	0.7270	-
	$K_{\text{Na}/\text{Mg}}$	$[0.1, 0.9]$	0.6965	-
	$K_{\text{Na}/\text{H}}$	$[1.0, 8.0] \times 10^{-6}$	1.69×10^{-6}	-
	pH_0	$[6.5, 9.0]$	6.9813	-
Initial concentration	C_{Ca}	$[1.0, 9.0] \times 10^{-3}$	3.32×10^{-3}	mol/L
	C_{Mg}	$[7.0, 40.0] \times 10^{-3}$	0.0121	mol/L
	C_{Na}	$[5.0, 10.0] \times 10^{-2}$	0.0938	mol/L
	C_{K}	$[1.0, 4.0] \times 10^{-3}$	1.63×10^{-3}	mol/L
	C_{HCO_3}	$[8.0, 40.0] \times 10^{-3}$	0.0121	mol/L
	pH_b	$[6.5, 9.0]$	7.8403	-
Boundary concentration	$\text{C}_{\text{Ca},b}$	$[0.8, 4.0] \times 10^{-3}$	2.03×10^{-3}	mol/L
	$\text{C}_{\text{Mg},b}$	$[0.8, 4.0] \times 10^{-6}$	3.64×10^{-6}	mol/L
	$\text{C}_{\text{Na},b}$	$[0.7, 4.0] \times 10^{-4}$	7.22×10^{-5}	mol/L
	$\text{C}_{\text{K},b}$	$[2.0, 9.0] \times 10^{-5}$	6.84×10^{-5}	mol/L
	$\text{C}_{\text{HCO}_3,b}$	$[1.0, 7.0] \times 10^{-3}$	3.61×10^{-3}	mol/L

goodness-of-fit is achieved through mechanistic causality rather than statistical overfitting.

Despite these advancements, the inversion results still exhibit notable parameter equifinality (Table 6), a phenomenon also observed by Dai and Samper (2006) and Chen et al. (2021). Different combinations of the 18 parameters can produce similarly acceptable concentration profiles along the flow path. This non-uniqueness is not eliminated by automation alone and should be interpreted as a limitation specific to the Hydro-Agent framework. Rather, it reflects the intrinsic ill-posedness of high-dimensional reactive transport inversion under limited field observations and the simplifying assumption of spatial homogeneity in the conceptual model. For this field-scale application, the observed equifinality is further compounded by multi-sourced real-world observational noise, making the decoupling of model structural errors from measurement uncertainty inherently challenging.

4.4. Synthesis of framework performance and practical implications

4.4.1. Computational efficiency, transparency, and optimization strategy

A primary motivation for introducing LLM-driven agents into hydrogeology is to shift the researcher's focus from low-level implementation details to high-level scientific reasoning. In our field-scale application (Case III), the Hydro-Agent autonomously executed the majority of the workflow, from script generation to iterative convergence, within approximately 5 h. In contrast, traditional expert-driven calibration for a comparable 18-parameter reactive transport model typically spans days to weeks, necessitating repeated cycles of manual script debugging and file formatting. This efficiency stems from the agent's emergent capacity for defensive programming. Detailed analysis of the execution logs reveals that the agent did not merely react to runtime crashes; it proactively implemented a robust validation layer, including type-checking and pre-optimization dry-runs, to isolate API mismatches. By reducing the effective downtime to near zero, the agent effectively decouples hypothesis testing from routine syntax logistics.

Crucially, this autonomy does not come at the cost of interpretability. Unlike black-box data-driven surrogates that obscure hydrological knowledge within opaque weight matrices, the Hydro-Agent operates on a "code-as-policy" paradigm. The framework maintains a fully transparent execution audit trail, documenting the decision-making rationale directly within the generated Python scripts. For regulatory compliance and engineering review, this transparency is indispensable. In the Aquia model, the self-documenting code allowed for direct verification of the Gaines-Thomas formulations and mass-balance constraints via the underlying numerical engine. Consequently, the resulting predictions are not merely statistically plausible artifacts, but mechanistically validated outcomes anchored in rigorous thermodynamic laws.

The performance of the Hydro-Agent framework is underpinned by its transition from a rigid execution interface to an adaptive optimization strategist. Analysis of the runtime logs (Table S5) demonstrates that the framework employs DE not as a terminal solver, but as a global explorer that systematically triggers high-precision local refinements based on the mathematical topology of the objective function. In Case I, the agent diagnosed the continuous gradients inherent in the steady-state hydraulic head distribution and autonomously invoked a gradient-based polish phase (L-BFGS-B), ensuring convergence beyond the initial stochastic search limits. In the more complex reactive transport scenario (Case II), the framework exhibited execution-aware resource management: to adhere to a strict 2-hour latency constraint without sacrificing fidelity, it dynamically reduced the population size (20 to 15) while implementing successive search-space contraction. Upon identifying numerical plateaus, the agent re-centered the parameter bounds to a narrow $\pm 2\%$ domain and transitioned to a TNC solver, effectively driving the residuals below the 5×10^{-5} precision threshold.

The robustness of the framework's autonomous engineering is further validated in the high-dimensional Aquia Aquifer model (Case

III). Addressing the statistical inefficiency of random sampling in an 18-parameter space, the agent replaced standard initialization with perturbed heuristic seeding derived from localized sensitivity insights. To safeguard against computational interruptions during the 100,000-year geochemical simulations, the framework engineered a persistence layer utilizing automated state-checkpoints. Furthermore, by adopting a deferred update protocol, the agent successfully minimized the computational overhead associated with the stiff multicomponent reaction network. These case-specific interventions, integrating solver hybridization, dynamic boundary adjustments, and real-time resource balancing, confirm that the Hydro-Agent can reconcile intensive search requirements with practical computational budgets, emulating the strategic decision-making of expert human modelers in complex inverse problems.

To further isolate the specific impact of LLM mediation, comparative testing against a directly coupled implementation (without LLM mediation) yielded statistically indistinguishable convergence levels and final residual norms. This confirms that the Hydro-Agent does not alter the mathematical structure of the inversion problem nor artificially improve the underlying optimizer's performance. Instead, its contribution lies in automating model configuration, optimization orchestration, and execution diagnostics while maintaining full adherence to thermodynamic constraints. In this sense, the framework improves operational efficiency and reproducibility rather than fundamentally redefining parameter identifiability limits.

4.4.2. Human-on-the-loop supervision and the boundary of geological reasoning

Despite the demonstrated operational autonomy, our results delineate a clear technical boundary: LLMs cannot yet substitute for the diagnostic capacity of a hydrogeologist in interpreting the mechanistic implications of modeling discrepancies. Case III revealed a critical divergence between mathematical optimality and geochemical consistency. When initially driven by a generic minimization objective, the agent successfully reduced the total residuals, but the resulting profiles were geochemically unsatisfactory; dominant species (e.g., Na^+) numerically masked errors in trace but kinetically significant components such as H^+ .

This limitation underscores a fundamental distinction between syntactic automation and semantic valuation. The Hydro-Agent demonstrated high proficiency in the computational architecture of modeling—efficiently generating simulator-compatible code, orchestrating global optimization routines, and debugging interface errors. However, it lacked the domain-specific heuristic to recognize that a subtle deviation in pH is more consequential for reactive transport kinetics than a proportional deviation in major cations. Upon observing these non-physical numerical artifacts after the initial simulation batch, human intervention was required to re-guide the agent's reasoning. By refining the system to mandate an inverse standard-deviation weighted RMSE, the expert effectively injected the necessary domain constraints to normalize the residuals across multiple scales. This synergy ensures that domain expertise is leveraged to formulate the mechanistic criteria, while the agent autonomously orchestrates the adaptive numerical strategies required to navigate the complex solution manifold.

These findings clarify how human supervision is integrated within the closed-loop workflow. Unlike traditional manual calibration, the human expert does not interrupt the computational process to run individual simulations. Instead, the expert provides high-level guidance when necessary. For example, when reliance on standard RMSE led to geochemically inconsistent solutions, the objective function was reformulated to better reflect the physical significance of specific components. Such interventions adjust the calibration criteria while preserving the computational momentum of the automated workflow. In this configuration, the framework performs the iterative search process, whereas the human supervisor ensures that convergence remains consistent with hydrogeochemical mechanisms. This division of roles

maintains computational efficiency while retaining scientific oversight. Furthermore, this expert oversight is essential for distinguishing between physical equifinality and numerical artifacts induced by observational noise, particularly in field-scale applications like Case III where real-world data uncertainty can synergistically mask true parameter identifiability.

Beyond parameter-level calibration, the current implementation operates within a predefined conceptual model. It does not independently revise stratigraphic interpretations, modify facies distributions, or adjust boundary conditions based on site-history information. Conceptual model development and hypothesis refinement therefore remain dependent on expert geological judgment. While the framework automates numerical implementation, it does not replace field-based conceptual reasoning required for complex site characterization under substantial epistemic uncertainty.

Several structural limitations should also be acknowledged. Although spatial heterogeneity in hydraulic conductivity was reconstructed in Case I, the coupled hydrogeochemical models in Cases II and III assumed spatially homogeneous reaction parameters. The framework did not invert spatially distributed geochemical reaction fields. In natural aquifer systems, mineralogical variability and facies-dependent exchange properties may introduce additional parameter heterogeneity and identifiability challenges. Extending the framework to accommodate spatially heterogeneous reaction parameterization represents an important direction for future work.

5. Conclusion

This study presents and rigorously validates Hydro-Agent, a physics-integrated framework that leverages LLMs as autonomous reasoning engines for groundwater inverse modeling. By establishing a code-as-policy paradigm, the framework bridges the gap between conceptual hypothesis formulation and rigorous numerical implementation, translating modeling intents into executable, auditable scripts. Our hierarchical validation, encompassing the structural identification of non-convex geological features (Case I), the high-precision resolution of non-linear competitive cation exchange kinetics across various stochastic noise levels (Case II), and the high-dimensional reactive transport calibration of the Aquia Aquifer (Case III), demonstrates that LLM-driven agents can effectively navigate rugged optimization landscapes. Crucially, the execution logs reveal that the agent contributes beyond purely numerical minimization by enabling workflow-level automation and physics-consistent execution: it demonstrated task-level reasoning reflected in its configuration choices and implemented defensive coding practices to preemptively handle numerical instabilities, ensuring that the inverted parameters are not only mathematically optimal but physically and syntactically robust.

Beyond algorithmic efficacy, the field-scale application reveals the emergence of a pivotal human-on-the-loop collaborative paradigm. The agent demonstrated operational capabilities analogous to a senior research engineer, autonomously formulating production-ready deployment protocols and rigorous error-handling mechanisms suitable for real-time monitoring. However, the optimization challenges encountered in the Aquia case underscore the indispensable role of domain expertise. Our findings indicate that while AI agents excel at syntactic implementation and operational logistics, they rely on human supervision to provide semantic valuation, such as defining weighting schemes necessary for geochemical consistency. Ultimately, Hydro-Agent acts as a force multiplier, enabling hydrogeologists to shift their focus from routine scripting to high-level conceptual synthesis.

Future development will extend the agent framework beyond parameter calibration to support automated geological model construction and numerical model setup. In particular, interfaces for stratigraphic geometry construction, boundary delineation, and mesh discretization will be integrated so that conceptual geological descriptions can be translated into simulation-ready computational models

within the agent workflow. In addition, future extensions will relax the current assumption of spatially homogeneous reaction parameters, allowing the framework to incorporate spatially distributed reaction parameter fields in heterogeneous aquifer systems. Furthermore, integrating multi-modal inputs and retrieval-augmented reasoning will allow the framework to incorporate diverse data sources such as borehole logs and cross-sections, enabling a more complete end-to-end hydrogeological modeling workflow.

Data availability

The source code supporting this study, along with the complete execution logs for Case Studies I-III, are openly available in the Hydro-Agent-Inversion repository on GitHub at <https://github.com/FuninGma-QUT/Hydro-Agent-Inversion>.

CRediT authorship contribution statement

Funing Ma: Writing – original draft, Methodology, Formal analysis, Conceptualization. **Junjun Chen:** Writing – review & editing, Investigation. **Zhenxue Dai:** Writing – review & editing, Funding acquisition. **Fangfei Cai:** Writing – review & editing. **Yingtao Hu:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.watres.2026.125886](https://doi.org/10.1016/j.watres.2026.125886).

References

- Agyei, E., Hatfield, K., 2006. Enhancing gradient-based parameter estimation with an evolutionary approach. *J. Hydrol.* 316 (1–4), 266–280.
- Appelo, C., 1994. Cation and proton exchange, pH variations, and carbonate reactions in a freshening aquifer. *Water Resour. Res.* 30 (10), 2793–2805.
- Bakker, M., et al., 2016. Scripting MODFLOW model development using Python and FloPy. *Groundwater* 54 (5), 733–739.
- Chen, J., Dai, Z., 2024. Metaheuristic algorithms for groundwater model parameter inversion: advances and prospects. *Deep Resour. Eng.* 1 (2), 100009.
- Chen, J., et al., 2021. An improved tandem neural network architecture for inverse modeling of multicomponent reactive transport in porous media. *Water Resour. Res.* 57 (12), e2021WR030595.
- Chen, J., Dai, Z., Yin, S., Zhang, M., Soltanian, M.R., 2025. Enhancing inverse modeling in groundwater systems through machine learning: a comprehensive comparative study. *Hydrol. Earth Syst. Sci.* 29 (17), 4251–4279.
- Chen, X., Lin, M., Schärli, N., Zhou, D., 2023. Teaching large language models to self-debug. *arXiv preprint arXiv:2304.05128*.
- Chen, Y., Jhamtani, H., Sharma, S., Fan, C., Wang, C., 2024. Steering large language models between code execution and textual reasoning. *arXiv preprint arXiv:2410.03524*.
- Dai, Z., Samper, J., 2004. Inverse problem of multicomponent reactive chemical transport in porous media: formulation and applications. *Water Resour. Res.* 40 (7).
- Dai, Z., Samper, J., 2006. Inverse modeling of water flow and multicomponent reactive transport in coastal aquifer systems. *J. Hydrol.* 327 (3–4), 447–461.
- Dai, Z., Zhan, C., Soltanian, M.R., Ritz, R.W., Zhang, X., 2019. Identifying spatial correlation structure of multimodal permeability in hierarchical media with Markov chain approach. *J. Hydrol.* 568, 703–715.
- Dai, Z., et al., 2025. Incorporating deep learning into hydrogeological modeling: advancements, challenges, and future directions. *J. Geophys. Res.: Mach. Learn. Comput.* 2 (2), e2025JH000703.

- Goldstein, Y., Perelman, G., Schuster, A., Ostfeld, A., 2025. Large language models for water distribution systems modeling and decision-making. *World Environ. Water Resour. Congr.* 2025 921–928.
- Hadid, A., Chakraborty, T., Busby, D., 2024. When geoscience meets generative AI and large language models: foundations, trends, and future challenges. *Expert Syst.* 41 (10), e13654.
- Hedaiaty Marzouny, N., Dziedzic, R., 2024. AI-assisted pump operation for energy-efficient water distribution systems. *Eng. Proc.* 69 (1), 3.
- Hill, M.C., Tiedeman, C.R., 2007. Effective Groundwater Model calibration: with Analysis of data, sensitivities, predictions, and Uncertainty. John Wiley & Sons.
- Hosseini, S.H., Pourzangbar, A., 2025. How well do DeepSeek, ChatGPT, and Gemini respond to water science questions? *Environ. Model. Softw.*, 106772.
- Hu, S. et al., 2025. Evaluating hydro-science and engineering knowledge of large language models. *arXiv preprint arXiv:2512.03672*.
- Huang, C., Dai, Z., 2008. Modeling groundwater in multimodal porous media with localized decompositions. *Math. Geosci.*
- Ji, Z., et al., 2023. Survey of hallucination in natural language generation. *ACM Comput. Surv.* 55 (12), 1–38.
- Karniadakis, G.E., et al., 2021. Physics-informed machine learning. *Nat. Rev. Phys.* 3 (6), 422–440.
- Ketabchi, H., Ataie-Ashtiani, B., 2015. Coastal groundwater optimization—Advances, challenges, and practical solutions. *Hydrogeol. J.* 23 (6), 1129–1154.
- Laloy, E., Vrugt, J.A., 2012. High-dimensional posterior exploration of hydrologic models using multiple-try DREAM (ZS) and high-performance computing. *Water Resour. Res.* 48 (1).
- Liang, J. et al., 2022. Code as policies: language model programs for embodied control. *arXiv preprint arXiv:2209.07753*.
- Liu, A. et al., 2024. Deepseek-v3 technical report. *arXiv preprint arXiv:2412.19437*.
- Liu, X., Song, Y., 2025. Scientific machine learning of flow resistance using universal shallow water equations with differentiable programming. *Water Resour. Res.* 61 (9), e2025WR040265.
- Lyu, H., Zhou, S.a., Wang, Z., Fu, G., Zhang, C., 2025. Assessing large multimodal models for urban floodwater depth estimation. *Water Resour. Res.* 61 (4), e2024WR039494.
- Ma, F., et al., 2025a. Impact of groundwater extraction intensity on the monitoring design for seawater intrusion in heterogeneous coastal aquifers. *J. Hydrol.*, 133638.
- Ma, Z., et al., 2025b. Embracing large language model (LLM) technologies in hydrology research. *Environ. Res.: Water* 1 (2), 022001.
- Ma, F., Cai, F., Zhang, L., Jia, S., 2026. Experimental and numerical investigation of colloid-facilitated strontium transport in fractured granite under different hydrogeochemical conditions[J]. *J. Contam. Hydrol.* 277, 104858.
- Mao, X.-m., et al., 2006. Three-dimensional model for multi-component reactive transport with variable density groundwater flow. *Environ. Model. Softw.* 21 (5), 615–628.
- Ng, C.W.W., Usman, M., Guo, H., 2023. Spatiotemporal pore-water pressure prediction using multi-input long short-term memory. *Eng. Geol.* 322, 107194.
- Pan, Z., Lu, W., Wang, H., Bai, Y., 2022. Fast inverse estimation of hydraulic conductivity field based on a deep convolutional-cycle generative adversarial neural network. *J. Hydrol.* 613, 128420.
- Raissi, M., Perdikaris, P., Karniadakis, G.E., 2019. Physics-informed neural networks: a deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *J. Comput. Phys.* 378, 686–707.
- Renard, P., 2007. Stochastic hydrogeology: what professionals really need? *Groundwater* 45 (5), 531–541.
- Samper, F.J., Neuman, S.P., 1989. Estimation of spatial covariance structures by adjoint state maximum likelihood cross validation: 1. Theory. *Water Resour. Res.* 25 (3), 351–362.
- Shinn, N., Cassano, F., Gopinath, A., Narasimhan, K., Yao, S., 2023. Reflexion: language agents with verbal reinforcement learning. *Adv. Neural Inf. Process. Syst.* 36, 8634–8652.
- Vrugt, J.A., 2016. Markov chain Monte Carlo simulation using the DREAM software package: theory, concepts, and MATLAB implementation. *Environ. Model. Softw.* 75, 273–316.
- Wang, J., Fu, G., Savic, D., 2025. Leveraging large language models for automating water distribution network optimization. *Water Res.*, 124536.
- Wang, L., et al., 2024. A survey on large language model based autonomous agents. *Front. Comput. Sci.* 18 (6), 186345.
- Wu, Q., et al., 2024. Autogen: enabling next-gen LLM applications via multi-agent conversations. In: *First Conference on Language Modeling*.
- Xia, X., Jiang, S., Zhou, N., Cui, J., Li, X., 2023. Groundwater contamination source identification and high-dimensional parameter inversion using residual dense convolutional neural network. *J. Hydrol.* 617, 129013.
- Xu, T., Sonnenthal, E., Spycher, N., Pruess, K., 2006. TOUGHREACT—A simulation program for non-isothermal multiphase reactive geochemical transport in variably saturated geologic media: applications to geothermal injectivity and CO₂ geological sequestration. *Comput. Geosci.* 32 (2), 145–165.
- Ye, H., et al., 2024. Reevo: large language models as hyper-heuristics with reflective evolution. *Adv. Neural Inf. Process. Syst.* 37, 43571–43608.
- Zhao, Y., Guo, Q., Lu, C., Luo, J., 2022. High-dimensional groundwater flow inverse modeling by upscaled effective model on principal components. *Water Resour. Res.* 58 (7), e2022WR032610.
- Zhou, H., Gómez-Hernández, J.J., Li, L., 2014. Inverse methods in hydrogeology: evolution and recent trends. *Adv. Water Resour.* 63, 22–37.